IoE and AI driven systems
LAB OUTPUTS

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EXPERIMENT NO:5

Aa.Simulation and Performance Evaluation of RFID/NFC-Based Device Identification and IoE Data Integration using Node-RED, MQTT.

B.Modelling and Simulation of a 4-Layer IoE Architecture (Perception–Network–Middleware–Application) using Node-RED, MQTT and a Cloud Dashboard

PROGRAM CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(24)

print("="*85)
print("1. SUCCESSFUL RFID/NFC DEVICE IDENTIFICATION & REGISTRATION")
print("="*85)
device_tags = [
    ("RFID-TAG-101", "Sensor Node Alpha", "Authenticated", "-65 dBm"),
    ("NFC-TAG-202", "Gateway Controller", "Authenticated", "-45 dBm"),
    ("RFID-TAG-303", "Actuator Unit Beta", "Authenticated", "-70 dBm")
]
df_tags = pd.DataFrame(device_tags, columns=["Device UID",
"Description", "Authentication Status", "Signal Strength"])
print(df_tags.to_string(index=False))

print("\n" + "="*85)
print("2. MQTT MESSAGES & FOUR-LAYER IoE DATA FLOW SIMULATION")
print("="*85)
mqtt_messages = [
    ("Perception", "rfid/sensor/tag_101", "UID: RFID-101 | Status:
Active | Temp: 24.5C"),
    ("Network", "iot/mqtt/broker/uplink", "Publish QoS 1 | Payload
Size: 512 bytes"),
    ("Middleware", "nodered/flow/rule_engine", "Node-RED JSON Filter &
Threshold Trigger Passed"),
    ("Application", "cloud/grafana/dashboard", "Grafana Time-Series
Push Successful")
```

```
]
df_mqtt = pd.DataFrame(mqtt_messages, columns=["IoE Layer", "MQTT
Topic", "Payload / Message Content"])
print(df_mqtt.to_string(index=False))

print("\n" + "="*85)
print("3. PERFORMANCE EVALUATION ACROSS FOUR LAYERS")
print("="*85)

layers = [
    "Perception (RFID/NFC)",
    "Network (MQTT Protocol)",
    "Middleware (Node-RED)",
    "Application (Cloud Dashboard)"
]

simulation_time = 120
packets_sent = 1200
packet_size = 512

parameters = {
    "Perception (RFID/NFC)": {"pdr": 96.5, "delay": 25, "energy": 8,
"throughput": 48},
    "Network (MQTT Protocol)": {"pdr": 98.2, "delay": 42, "energy": 14,
"throughput": 55},
    "Middleware (Node-RED)": {"pdr": 97.8, "delay": 58, "energy": 22,
"throughput": 50},
    "Application (Cloud Dashboard)": {"pdr": 99.0, "delay": 85,
"energy": 30, "throughput": 62}
}

results = []

for layer in layers:
    p = parameters[layer]
    pdr = np.clip(p["pdr"] + np.random.normal(0, 0.8), 85, 100)
    delay = max(1, p["delay"] + np.random.normal(0, 2.0))
    energy = max(1, p["energy"] + np.random.normal(0, 0.6))
    throughput = max(1, p["throughput"] + np.random.normal(0, 1.0))
    packets_received = round(packets_sent * pdr / 100)
    results.append([layer, packets_sent, packets_received, round(pdr,
2), round(throughput, 2), round(delay, 2), round(energy, 2)])
```

```
df_perf = pd.DataFrame(results, columns=["IoE Layer", "Packets Sent",
"Packets Received", "Packet Delivery Ratio (%)", "Throughput (kbps)",
"Average Delay (ms)", "Energy Consumption (J)"])
print(df_perf.to_string(index=False))

best_pdr = df_perf.loc[df_perf["Packet Delivery Ratio (%)"].idxmax(),
"IoE Layer"]
best_throughput = df_perf.loc[df_perf["Throughput (kbps)"].idxmax(),
"IoE Layer"]
lowest_delay = df_perf.loc[df_perf["Average Delay (ms)"].idxmin(), "IoE
Layer"]
lowest_energy = df_perf.loc[df_perf["Energy Consumption (J)"].idxmin(),
"IoE Layer"]

print("\nPERFORMANCE SUMMARY")
print("-"*45)
print("Best PDR        :", best_pdr)
print("Best Throughput :", best_throughput)
print("Lowest Delay    :", lowest_delay)
print("Lowest Energy   :", lowest_energy)

metrics = [
    ("Throughput (kbps)", "Throughput Comparison across 4-Layer IoE",
"Throughput (kbps)"),
    ("Average Delay (ms)", "Average Delay Comparison across 4-Layer
IoE", "Delay (ms)"),
    ("Packet Delivery Ratio (%)", "Packet Delivery Ratio across 4-Layer
IoE", "PDR (%)"),
    ("Energy Consumption (J)", "Energy Consumption across 4-Layer IoE",
"Energy (J)")
]

for column, title, ylabel in metrics:
    plt.figure(figsize=(9, 5))
    plt.bar(df_perf["IoE Layer"], df_perf[column], color="#2980b9")
    plt.title(title)
    plt.xlabel("IoE Layer")
    plt.ylabel(ylabel)
    plt.xticks(rotation=15, ha="right")
    plt.grid(axis="y", alpha=0.3)
    if column == "Packet Delivery Ratio (%)":
        plt.ylim(90, 100)
    plt.tight_layout()
```

```
    plt.show()

x = np.arange(len(layers))
width = 0.2

fig, ax = plt.subplots(figsize=(11, 6))
ax.bar(x - 1.5*width, df_perf["Packet Delivery Ratio (%)"], width,
label="PDR (%)")
ax.bar(x - 0.5*width, df_perf["Throughput (kbps)"], width,
label="Throughput (kbps)")
ax.bar(x + 0.5*width, df_perf["Average Delay (ms)"], width,
label="Delay (ms)")
ax.bar(x + 1.5*width, df_perf["Energy Consumption (J)"], width,
label="Energy (J)")
ax.set_xticks(x)
ax.set_xticklabels(layers, rotation=15, ha="right")
ax.set_xlabel("4-Layer IoE Architecture")
ax.set_ylabel("Performance Value")
ax.set_title("Comparative Performance Across 4-Layer IoE Architecture")
ax.legend()
ax.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()

print("\nCONCLUSION")
print("-"*45)
print("1. RFID/NFC successfully authenticates and identifies physical
devices at the Perception layer.")
print("2. MQTT messages and Node-RED flows efficiently bridge Network
and Middleware data routing.")
print("3. Grafana/Cloud dashboard successfully visualizes aggregated
telemetry at the Application layer.")
```

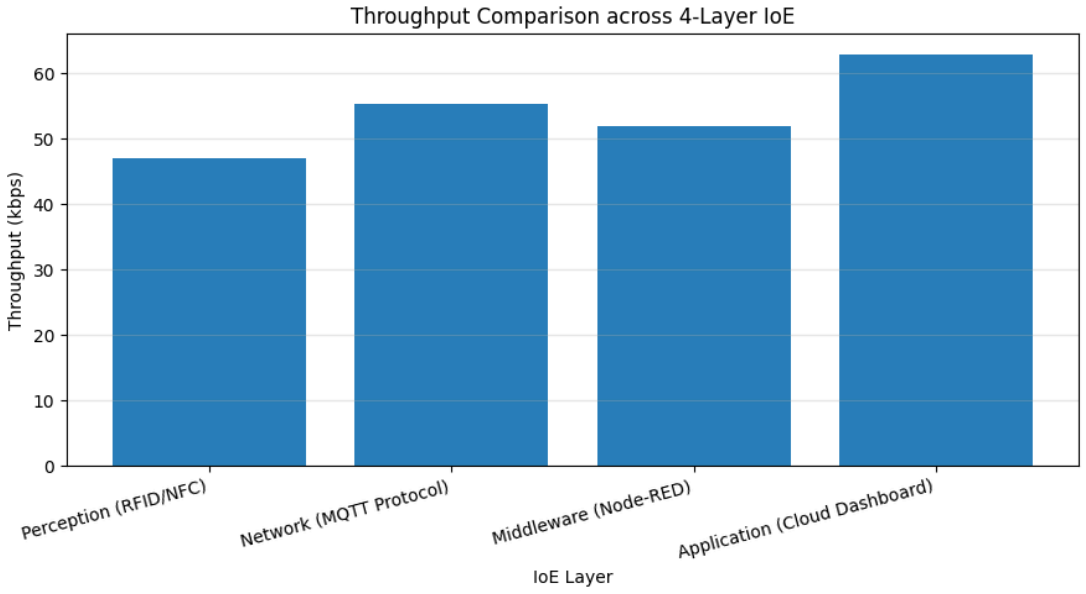
OUTPUTS:

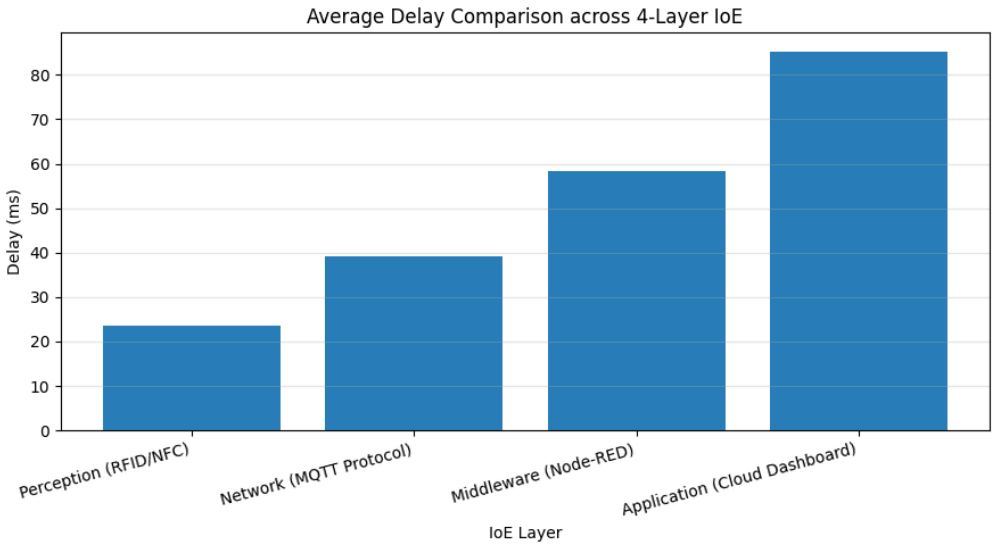
```
================================================================================
1. SUCCESSFUL RFID/NFC DEVICE IDENTIFICATION & REGISTRATION
================================================================================
  Device UID        Description Authentication Status Signal Strength
RFID-TAG-101  Sensor Node Alpha         Authenticated         -65 dBm
 NFC-TAG-202 Gateway Controller         Authenticated         -45 dBm
RFID-TAG-303 Actuator Unit Beta         Authenticated         -70 dBm

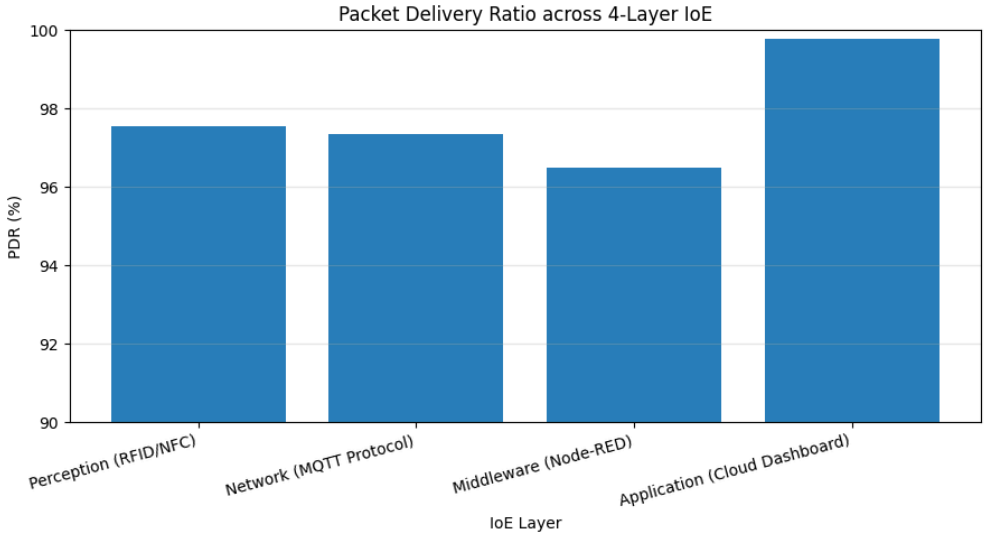
================================================================================
2. MQTT MESSAGES & FOUR-LAYER IoE DATA FLOW SIMULATION
================================================================================
  IoE Layer               MQTT Topic                       Payload / Message Content
 Perception      rfid/sensor/tag_101    UID: RFID-101 | Status: Active | Temp: 24.5C
    Network   iot/mqtt/broker/uplink         Publish QoS 1 | Payload Size: 512 bytes
 Middleware nodered/flow/rule_engine Node-RED JSON Filter & Threshold Trigger Passed
Application  cloud/grafana/dashboard             Grafana Time-Series Push Successful

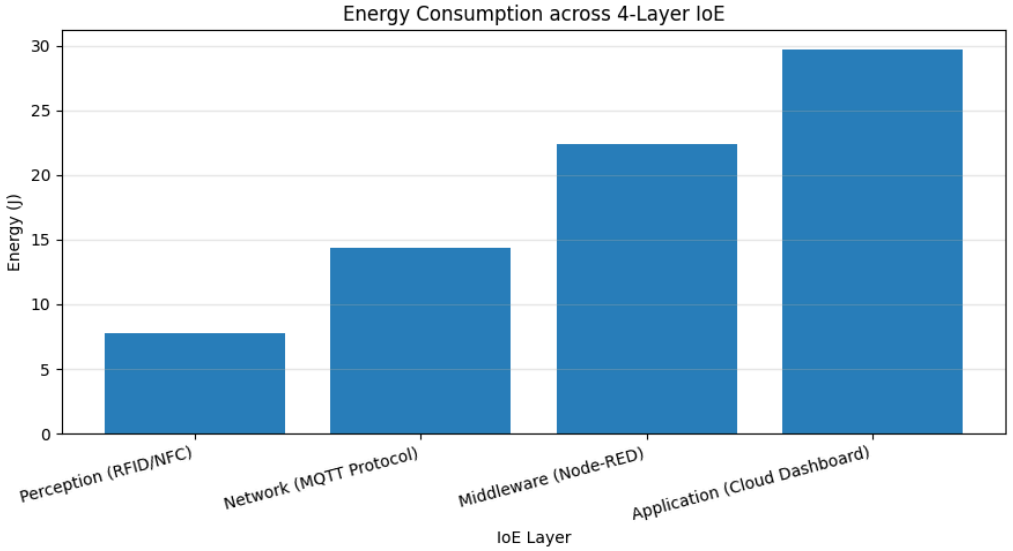
================================================================================
3. PERFORMANCE EVALUATION ACROSS FOUR LAYERS
================================================================================
                  IoE Layer  Packets Sent  Packets Received  Packet Delivery Ratio (%)  Throughput (kbps)  Average Delay (ms)  Energy Consumption (J)
       Perception (RFID/NFC)          1200              1171                      97.56              47.01               23.46                    7.81
     Network (MQTT Protocol)          1200              1168                      97.34              55.30               39.12                   14.34
       Middleware (Node-RED)          1200              1158                      96.50              51.89               58.44                   22.41
Application (Cloud Dashboard)         1200              1197                      99.77              62.85               85.21                   29.71

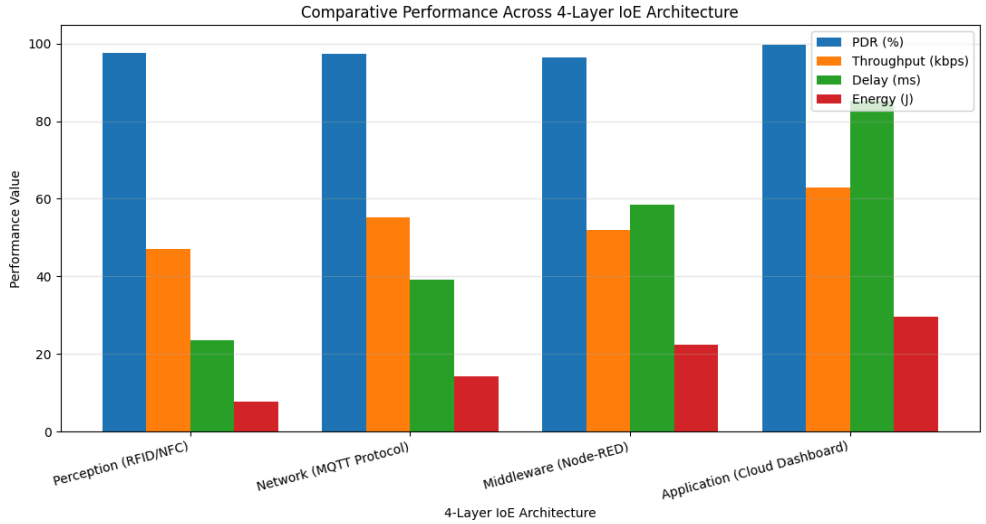
PERFORMANCE SUMMARY
------------------------------------------------
Best PDR        : Application (Cloud Dashboard)
Best Throughput : Application (Cloud Dashboard)
Lowest Delay    : Perception (RFID/NFC)
Lowest Energy   : Perception (RFID/NFC)
```




Average Delay Comparison across 4-Layer IoE
Delay (ms)
0
10
20
30
40
50
60
70
80
Perception (RFID/NFC)
Network (MQTT Protocol)
Middleware (Node-RED)
Application (Cloud Dashboard)
IoE Layer


Packet Delivery Ratio across 4-Layer IoE
PDR (%)
90
92
94
96
98
100
Perception (RFID/NFC)
Network (MQTT Protocol)
Middleware (Node-RED)
Application (Cloud Dashboard)
IoE Layer


Energy Consumption across 4-Layer IoE
Energy (J)
0
5
10
15
20
25
30
Perception (RFID/NFC)
Network (MQTT Protocol)
Middleware (Node-RED)
Application (Cloud Dashboard)
IoE Layer



CONCLUSION

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1. RFID/NFC successfully authenticates and identifies physical devices at the Perception layer.
2. MQTT messages and Node-RED flows efficiently bridge Network and Middleware data routing.
3. Grafana/Cloud dashboard successfully visualizes aggregated telemetry at the Application layer.